

Kishore R M

Software Development Engineer

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Summary

Results-driven Software Development Engineer with a master's degree and 4.5+ years of experience in developing scalable web applications and integrating complex systems. Expert in Agile Software Development methodologies and Object-Oriented Design with a proven track record in enhancing user experience, optimizing TypeScript and JavaScript frameworks, and improving system performance. Skilled in creating intuitive interfaces and leveraging emerging technologies to solve complex problems.

Experience

Singular Analysts Inc, USA | Senior Software Engineer

Sep 2024 – Present

- Spearheaded the integration of **GraphQL APIs** into a **Spring Boot** backend, enabling flexible data retrieval and reducing over-fetching in frontend requests.
- Developed RESTful microservices using **Java 8, Spring Boot, and Spring Cloud** for high-throughput airline transaction workflows.
- Built event-driven systems using **Kafka** producers and consumers for real-time flight status updates and booking confirmations.
- Optimized microservice performance using **Redis for caching** frequent seat availability queries and user session tokens.
- Containerized applications using **Docker** and deployed microservices to **Kubernetes clusters with Helm charts**.
- Architected a feature-flag system for backend modules using **AWS AppConfig** and **Lambda layers**, allowing dynamic control of rollout strategies without redeployment.
- Led the creation of self-documenting internal APIs using **SpringDoc** and **Kotlin DSL**, improving developer onboarding time and reducing integration errors across teams.
- Restructured data models in **PostgreSQL** and **MongoDB** to support analytical workloads and **multi-tenant isolation**, leveraging **materialized views, partial indexes, and schema versioning** to boost query performance and maintain scalability.
- Built an integrated testing strategy using **JUnit, Postman collections, and React Testing Library**, automating end-to-end and component-level validation across microservices and UI layers, and embedding tests into the **CI/CD pipeline** to enforce high code quality and prevent regressions.
- Implemented service discovery and load balancing with **Spring Cloud Eureka** and **Ribbon**. Tuned consumer groups for optimal parallel processing and lag monitoring.
- Implemented **JPA/Hibernate** with native queries for performance-critical endpoints. **Built GitHub Actions** pipelines for CI/CD automation with Maven builds, JUnit tests, and SonarQube checks. Integrated **Dynatrace** and **CloudWatch** for proactive health monitoring and alerting in production.
- Built a design system in **React** with reusable components, accessibility hooks, and theming support using **Tailwind CSS** and **Headless UI**, accelerating frontend delivery and maintaining brand consistency.
- Implemented **component-level** performance profiling in **React** using **React Profiler** and custom metrics sent to **CloudWatch**, identifying bottlenecks and reducing render time by 35%.
- Designed a **plugin-ready micro frontend architecture** using **Webpack Module Federation** and **React**, enabling independent team deployments and runtime integration of client-specific features.
- Actively participated in **Sprint Reviews, RCA meetings, and post-incident** playbook documentation.

Center for Technology in Government, USA | Full Stack Developer

Jan 2023 – Dec 2023

- Orchestrated the development of a secure, real-time **FEMA Message Design Dashboard (MDD)** using **ReactJS** and **Next.js 14**, improving emergency communication workflows and integrating testing requirements through **JIRA**.
- Integrated **API endpoints** with robust input **validation, exception handling, and retry mechanisms** across GET/POST/DELETE methods.
- Used **Kafka Connect** with Schema Registry to ensure **schema consistency** across multiple microservices. Wrote complex **SQL joins and triggers in PostgreSQL** for flight itinerary tracking and historical data.
- Configured readiness/liveness probes, auto-scaling policies, and resource quotas for high availability. Developed integration test scenarios using **Cucumber BDD** and automated them via **GitHub Actions** workflows.
- Performed root cause analysis using **Grafana** dashboards, **Splunk** logs, and thread dumps for memory leaks.
- Led backend development with **AWS Lambda** and **PostgreSQL (RDS)**, implementing reusable **Python** frameworks to improve system performance and reduce future development time.
- Integrated **AWS Amplify UI** with **Cognito User Pools** to streamline authentication and authorization, accelerating user onboarding and improving access control.
- Configured secure **AWS VPCs** and **IAM** policies, achieving government-level compliance while mitigating risks tied to infrastructure and data exposure.
- Automated infrastructure deployment via **Jenkins** pipelines and **CloudFormation**, enabling consistent **CI/CD** processes and reducing manual configuration errors.
- Integrated **AWS Amplify UI** with **Cognito User Pools** to streamline authentication and authorization, accelerating user onboarding and improving access control.
- Configured secure **AWS VPCs** and **IAM** policies, achieving government-level compliance while mitigating risks tied to infrastructure and data exposure.
- Ensured **WCAG 2.1** accessibility compliance by implementing screen-reader support, keyboard navigation, and responsive layouts using **Tailwind CSS**, improving usability for diverse user groups.
- Created comprehensive API documentation with **Swagger/OpenAPI**, expediting internal team onboarding and external integration readiness.
- Monitored application performance with **AWS CloudWatch**, integrating logs and custom metrics to enable proactive alerting and reduce incident response time.

ValueLabs LLC, India | Senior Software Engineer

Oct 2019 – Jul 2022

- Developed and deployed 50+ scalable RESTful APIs using **Spring Boot, AWS Lambda, and API Gateway**, improving data flow and storage by integrating seamlessly with **PostgreSQL** on **AWS RDS** to improve backend efficiency and reliability in data storage operations.
- Integrated **ReactJS** with **Spring Boot** through Object-Oriented Design principles resolving cross-layer compatibility challenges and enhancing scalability using **Docker-based microservices**.
- Leveraged **Docker** and **Kubernetes** for microservice containerization and orchestration, applying **DevOps** practices to improve system performance and scalability across distributed environments.
- Engineered secure authentication flows with **AWS Cognito** and **Two-Factor Authentication**, enhancing production reliability through **TDD** practices and ensuring robust compliance with security standards.
- Managed **hybrid** deployments across on-premises servers and **AWS Cloud** for 10,000+ users, streamlining delivery pipelines and reducing deployment friction by leveraging **AWS Elastic Beanstalk** for **legacy** systems.
- Refactored and optimized SQL queries for **PostgreSQL** and **MySQL** databases, applying algorithm design techniques to significantly reduce query latency and improve backend throughput under high-volume data operations.

Skills

Methodologies: Agile, SDLC, Waterfall, Scrum
Development Practices : CI/CD, TDD, GitOps, DevOps
Programming Languages : Java, Python, C, TypeScript, JavaScript (ES6+), SQL, Kotlin, Bash
Web Development : HTML5, CSS3, React.js, Next.js, Node.js, GraphQL, RESTful APIs, Tailwind CSS, Bootstrap, jQuery
Frameworks and Libraries : Spring Boot, Express.js, React Testing Library, SpringDoc, Headless UI, Webpack Module Federation, Redux
Build Tools, Platforms & Databases : AWS (Lambda, API Gateway, RDS, AppConfig, Step Functions, S3, CloudWatch, Cognito, VPC), Jenkins, Docker, Kubernetes, PostgreSQL, MongoDB, MySQL, DynamoDB, CloudFormation
Testing & Observability : JUnit, Postman, Swagger/OpenAPI, React Testing Library, AWS CloudWatch Logs Insights, Grafana
Tools & Environments : Jira, Git, GitHub, GitLab CI, Visual Studio Code, macOS, Linux, Windows
Certifications : AWS Certified Cloud Practitioner
Relevant Skills

Application Development, Automation, Collaboration, DevOps Practices, Architecture, Result Orientation, Solution Delivery Process, Analytical Thinking, Risk Management, Technical Strategy Development, CI/CD, Unit Testing, Integration Testing, Regression Testing, Performance Testing, Release Management, Story Refinement, Requirements Estimation, Branching Strategy, Pull Requests, Merge Conflict Resolution, Documentation, Code Review, Deployment Support.

Education

M.S., Data Science University at Albany, Albany, NY	Aug 2022 – May 2024
B. Tech., Electronics and Communications S R University, India	Jun 2016 – May 2020